

FRONTIER CLASS

Running GLM-5.2: locally and in the cloud

Only ~40B parameters are active per token, so the weights have to be held, not computed: a big-memory workstation at 4-bit locally, eight H200s at FP8 in the cloud. **Model:** GLM-5.2 · 744B MoE, ~40B active per token · MIT · 1M tokens · FP8 ≈ 744 GB · 4-bit GGUF 372–475 GB · 2-bit 239 GB.

744B

TOTAL PARAMS, MOE

40B

ACTIVE PER TOKEN

192 GB

VRAM, 2 × PRO 6000

512 GB

SYSTEM RAM, 8-CHANNEL

1M

CONTEXT TOKENS

LOCAL · BOUGHT ONCE**\$60,672**

The CUDA hybrid workstation as specced below; Newegg street prices on October 7, 2026.

CLOUD · HELD WARM**\$18,300**/mo

Northflank nf-gpu-h200-141-8g, 8 × H200 141 GB; list price, 24/7, CPU and RAM bundled.

01 Local — The CUDA hybrid workstation

Two Blackwell cards hold attention, the shared layers and the KV cache at CUDA speed with native FP8; 512 GB of 8-channel DDR5 holds the cold experts. Decode lands around 10–18 tok/s at 4-bit, prefill is CUDA-fast.

PART	PICK · WHY IT'S HERE	STREET	BUY
GPU × 2	NVIDIA RTX PRO 6000 Blackwell Max-Q, 96 GB × 2 Attention, dense layers, the FP8 KV cache and as many hot expert tensors as fit, split across both cards. Two Max-Q blowers at 300 W each sit side by side and run off a 1,600 W supply; the 600 W Workstation Edition pair would want 2,000 W and more airflow.	\$34,000 \$16,999.99 each in stock	Newegg ↗
CPU	AMD Ryzen Threadripper PRO 7975WX, 32 cores Eight DDR5 channels (~325 GB/s) feed the experts, which is the real bottleneck, plus 128 PCIe 5.0 lanes for two x16 cards and the offload engine.	\$3,800 in stock	Newegg ↗
MAINBOARD	ASUS Pro WS WRX90E-SAGE SE sTR5 / WRX90: seven PCIe 5.0 x16 slots, eight DDR5 RDIMM slots, 10 GbE. The only desktop platform with both the lanes and the memory channels.	\$1,205 in stock · list \$1,300	Newegg ↗
MEMORY	Kingston Server Premier 64 GB DDR5-5600 ECC RDIMM × 8 = 512 GB × 8 Holds the whole 4-bit GGUF (~475 GB) in RAM; experts stream to the GPUs per token. All eight channels populated: bandwidth sets decode speed. Memory prices roughly quadrupled since the June 2026 sheet; this line is where the money went.	\$18,800 \$2,349.99 each in stock	Newegg ↗
POWER	Seasonic PRIME TX-1600, 1600 W, 80+ Titanium 2 × 300 W Max-Q + 350 W CPU + platform ≈ 1.25 kW continuous. Native 12V-2x6 cables for both cards.	\$742 in stock	Newegg ↗
CPU COOLER	SilverStone XE360-TR5-V2, 360 mm AIO Full sTR5 cold-plate coverage for the 350 W Threadripper while two GPUs dump heat into the case.	\$386 in stock · list \$430	Newegg ↗
CASE	Fractal Design Meshify 2 XL Mesh airflow and room for two long cards and an E-ATX / SSI-EEB board. Alternative: a 4U rack chassis if it lives in a closet.	\$240 in stock	Newegg ↗
STORAGE	Samsung 9100 PRO 4 TB, NVMe Gen5 × 2 × 2 GGUF tiers run 223–810 GB each; Gen5 keeps mmap and load fast. Two 4 TB drives hold several quant levels side by side (the 8 TB model is not listed).	\$1,500 \$749.99 each in stock · list \$1,360	Newegg ↗
All-in, as specced · 8 parts, 8 priced live from newegg.com on October 7, 2026		\$60,672	

Newegg street prices as read on October 7, 2026; the live table at flatclaw.org/builds refreshes hourly. Used-market and unlisted parts are marked. Verify at checkout.

02 Alternatives Newegg has today (not in the total)

PART	PICK · WHEN TO TAKE IT	STREET	BUY
MEMORY · ALT NOT IN TOTAL	Kingston FURY Renegade Pro 128 GB (4 × 32 GB) DDR5-5600 RDIMM × 2 = 256 GB × 2 Half the capacity for the 2-bit and 3-bit tiers (239–360 GB with the VRAM); the cheaper way to stand it up and judge quality first.	\$9,600 \$4,799.99 each in stock	Newegg ↗

03 Cloud — the same checkpoint, one service

NORTHFLANK · RECOMMENDED · nf-gpu-h200-141-8g

\$18,300/mo · \$25/hr · spot \$14.6/hr ≈ \$10,500/mo

8 × H200 141 GB · 1,128 GB. Full-precision FP8 with KV room: production parity. SGLang --tp 8 --enable-moe-ep, context ≤256K. One service spanning eight GPUs; the weights-server + customEntrypoint pattern from the 31B box, with a ≥1.5 TB volume and a 20–40 minute cold boot (744 GB fetched).

NORTHFLANK · nf-gpu-b200-180-8g

\$34,000/mo · \$47/hr

8 × B200 180 GB · 1,536 GB. FP8 KV cache for the full 1M-token context.

NORTHFLANK · weights-server (CPU) + ≥1.5 TB volume

\$60/mo

none. nginx autoindex over the checkpoint directory; the GPU node fetches it at boot.

ALL-IN

~\$18,300 a month held warm on eight H200s, ~\$10,500 on spot; scale to zero between sessions.

THE SAME SHAPE ELSEWHERE

- ▶ **Azure** · ND H200 v5 (8 × H200) · ND GB200 v6 for 1M context — One 8-GPU deployment on AKS (NVIDIA device plugin, --tp 8) behind an internal ingress, or an Azure ML online endpoint; weights in Blob, fetched on boot; no public egress.
- ▶ **Azure, cheaper** · NC A100 v4 × N — Ampere has no FP8: a quantized GLM-5.2 only, the same caveat as the local box.

NOTES

- ▶ Cost reality: this is a different tier. Eight H200s are about ten times the 31B box held warm, so the frontier class is scheduled for live load and scaled to zero between demos.
- ▶ Per GPU-hour on Northflank: H100 \$2.74 · RTX PRO 6000 96 GB \$3.00 · H200 ~\$3.14 · B200 ~\$5.87.
- ▶ Routing is unchanged: OpenClaw points at one endpoint; a LiteLLM or vLLM router in the same tenancy adds failover to a quantized fallback or a second region.

04 The quant ladder — what fits where

TIER	FOOTPRINT	FITS
BF16 full precision	~1,500 GB	cloud only, multi-node
FP8 (the prod checkpoint)	~744 GB	8 × H200 cloud node
8-bit	810 GB	GPU server or cloud
5-bit	570 GB	512 GB RAM + 192 GB VRAM, tight
4-bit (UD-Q4_K_XL), chosen locally	372–475 GB	the 512 GB hybrid box, near-lossless
3-bit	290–360 GB	a 384 GB box
2-bit (UD-IQ2_M)	239 GB	a 256 GB box; ~82% top-1 at 84% smaller
1-bit	223 GB	tight budgets; the biggest quality hit

05 Run it

```
# Launch
./llama-server -m GLM-5.2-UD-Q4_K_XL.gguf -ngl 99 -n-cpu-moe 999 -ts 50,50 -c 131072 --flash-attn --cache-type-k q8_0 --cache-type-v q8_0 --host 0.0.0.0 --port 8000

# Alternative
ktransformers --model zai-org/GLM-5.2 --gguf_path ./GLM-5.2-UD-Q4 --cpu_infer 32 --max_new_tokens 131072 --port 8000
```

- ▶ The engine keeps attention and KV on the cards and routes cold experts to RAM: llama.cpp (-ngl 99 -n-cpu-moe 999 -ts 50,50 --flash-attn --cache-type-k q8_0 --cache-type-v q8_0) or KTransformers for more expert throughput. OpenClaw points at the single OpenAI endpoint, exactly like the 31B box.
- ▶ A 744B MoE owns all the memory: no room for image or voice on this box; run modality on a second machine as the standard build does.
- ▶ Honest line: locally you run a quantized GLM-5.2 (4-bit is excellent); FP8 production parity is the 8-GPU node's job.

06 The all-GPU 4U server (the cloud node, bought)

Eight 96 GB cards in a 4U chassis: 768 GB of VRAM, the full SGLang stack, 240 V in the closet. FP8's 744 GB fits with almost no KV room, so run 8-bit or 4-bit with headroom, or FP8 at short context.

PART	PICK · WHY IT'S HERE	STREET	BUY
CHASSIS	ASUS ESC8000A-E12, 4U 8-GPU barebone (dual EPYC 9004) Eight double-width GPU bays with the airflow and the redundant 3 kW power to match; the barebone includes board, backplane and supplies.	\$6,099 out of stock	Newegg ↗
CPU × 2	AMD EPYC 9354, 32 cores × 2 Two sockets for 24 memory channels and 256 PCIe 5.0 lanes; the GPUs do the work.	\$6,000 \$3,000.00 each not sold as a part on Newegg; ~\$3,000 each street	search ↗
MEMORY	Kingston Server Premier 64 GB DDR5-5600 ECC RDIMM × 12 = 768 GB × 12 Matches the VRAM so a full checkpoint can stage through RAM; one DIMM per channel across both sockets.	\$28,200 \$2,349.99 each in stock	Newegg ↗
GPU × 8	NVIDIA RTX PRO 6000 Blackwell Max-Q, 96 GB × 8 300 W blowers were made for dense multi-GPU boxes; 8 × 96 GB = 768 GB. The passive Server Edition is the other option when it is in stock at a sane price.	\$136,000 \$16,999.99 each in stock	Newegg ↗
STORAGE	Samsung 9100 PRO 4 TB, NVMe Gen5 × 4 × 4 The FP8 checkpoint alone is 744 GB; keep two quant tiers and the BF16 shards for conversion.	\$3,000 \$749.99 each in stock · list \$1,360	Newegg ↗
All-in, as specced · 5 parts, 4 priced live from newegg.com on October 7, 2026, the rest at reference prices		\$179,299	

Launch

```
python -m sglang.launch_server --model-path zai-org/GLM-5.2 --tp 8 --quantization fp8 --enable-moe-ep --kv-cache-dtype fp8_e5m2 --context-length 262144 --port 8000
```

- ▶ 8 × 300 W + platform ≈ 3 kW: a dedicated 240 V circuit and a room that can shed the heat.
- ▶ This is the same shape as the cloud node, which is the point: the launch line (sglang --tp 8 --quantization fp8) is identical, only the owner changes.